The Quartic Interaction Vertex

1 Quartic contributions from various terms of the Lagrangian

$$\boldsymbol{L_1 : e^{\psi}\left(2\partial_+\partial_-\psi - \frac{1}{2}\partial_+\gamma^{ij}\partial_-\gamma_{ij}\right)}$$

$$2\psi\partial_+\partial_-\psi + \psi[\partial_+ h\partial_-\bar{h} + \partial_- h\partial_+\bar{h}] + \frac{1}{3}[\partial_+(h h\bar{h})\partial_-\bar{h} + \partial_+(\bar{h}\bar{h}h)\partial_- h]$$

$$+\frac{1}{3}[\partial_-(h h\bar{h})\partial_+\bar{h} + \partial_-(\bar{h}\bar{h}h)\partial_+ h] - \partial_+(h\bar{h})\partial_-(h\bar{h})$$

$$\boldsymbol{L_2 : -e^{\frac{\psi}{2}}\gamma^{ij}\left(\frac{1}{2}\partial_i\partial_j\psi - \frac{3}{8}\partial_i\psi\partial_j\psi - \frac{1}{4}\partial_i\gamma^{kl}\partial_j\gamma_{kl} + \frac{1}{2}\partial_i\gamma^{kl}\partial_k\gamma_{jl}\right)}$$

$$-\frac{1}{2}\psi\partial\bar{\partial}\psi - h\bar{h}\partial\bar{\partial}\psi + \frac{3}{4}\partial\psi\bar{\partial}\psi - \frac{1}{2}\psi[\partial h\bar{\partial}\bar{h} + \partial\bar{h}\bar{\partial}h] + \psi\partial h\bar{\partial}\bar{h}$$

$$-\frac{1}{3}[\bar{\partial}(h h\bar{h})\partial\bar{h} + \bar{\partial}(\bar{h}\bar{h}h)\partial h + \partial(h h\bar{h})\bar{\partial}\bar{h} + \partial(\bar{h}\bar{h}h)\bar{\partial}h]$$

$$+\frac{2}{3}[\bar{\partial}(\bar{h}\bar{h}h)\partial h + \partial(h h\bar{h})\bar{\partial}\bar{h}] + \partial(h\bar{h})\bar{\partial}(h\bar{h}) - \partial(h\bar{h})\bar{\partial}(h\bar{h})$$

$$-h\bar{h}[\partial h\bar{\partial}\bar{h} + \partial\bar{h}\bar{\partial}h] + [h\partial\bar{h} - \bar{h}\partial h]\bar{\partial}(h\bar{h}) + [\bar{h}\bar{\partial}h - h\bar{\partial}\bar{h}]\partial(h\bar{h}) + 2h\bar{h}\partial h\bar{\partial}\bar{h}$$

$$\boldsymbol{L_3 : -\frac{1}{2}e^{-\frac{3\psi}{2}}\gamma^{ij}\frac{1}{\partial_-}R_i\frac{1}{\partial_-}R_j}$$

where

$$\boldsymbol{R_i = e^{\psi}(\frac{1}{2}\partial_-\gamma^{jk}\partial_i\gamma_{jk} - \frac{3}{2}\partial_-\partial_i\psi + \frac{1}{2}\partial_-\psi\partial_i\psi) + \partial_k(e^{\psi}\gamma^{jk}\partial_-\gamma_{ij})}$$

$$3\psi\partial\bar{h}\bar{\partial}h - \frac{3}{2}\bar{\partial}\psi\frac{1}{\partial_-}[\partial_-h\partial\bar{h} + \partial_-\bar{h}\partial h] - \frac{3}{2}\partial\psi\frac{1}{\partial_-}[\partial_-h\bar{\partial}\bar{h} + \partial_-\bar{h}\bar{\partial}h] - \frac{9}{4}\partial\psi\bar{\partial}\psi$$

$$-3\partial\psi\frac{\bar{\partial}}{\partial_-}[h\partial_-\bar{h}] - 3\bar{\partial}\psi\frac{\partial}{\partial_-}[\bar{h}\partial_-h] + \frac{3}{2}\partial\psi\bar{\partial}(h\bar{h}) + \frac{3}{2}\bar{\partial}\psi\partial(h\bar{h})$$

$$-3h\partial\bar{h}\bar{\partial}\psi - 3\bar{h}\bar{\partial}h\partial\psi - 2\partial\bar{h}\frac{\bar{\partial}}{\partial_-}[\psi\partial_-h] - 2\bar{\partial}h\frac{\partial}{\partial_-}[\psi\partial_-\bar{h}]$$

$$-2\frac{\partial}{\partial_-}[h\bar{h}\partial_-\bar{h}]\bar{\partial}h - 2\frac{\bar{\partial}}{\partial_-}[h\bar{h}\partial_-h]\partial\bar{h} + 2\frac{\partial}{\partial_-}[\bar{h}\partial_-(h\bar{h})]\bar{\partial}h + 2\frac{\bar{\partial}}{\partial_-}[h\partial_-(h\bar{h})]\partial\bar{h}$$

$$-2\frac{1}{\partial_-}[\partial_-h\bar{\partial}\bar{h} + \partial_-\bar{h}\bar{\partial}h]\frac{\partial}{\partial_-}[\bar{h}\partial_-h] - 2\frac{1}{\partial_-}[\partial_-h\partial\bar{h} + \partial_-\bar{h}\partial h]\frac{\bar{\partial}}{\partial_-}[h\partial_-\bar{h}]$$

$$-4\frac{\bar{\partial}}{\partial_-}[h\partial_-\bar{h}]\frac{\partial}{\partial_-}[\bar{h}\partial_-h] - \frac{1}{\partial_-}[\partial_-h\bar{\partial}\bar{h} + \partial_-\bar{h}\bar{\partial}h]\frac{1}{\partial_-}[\partial_-\bar{h}\partial h + \partial_-h\partial\bar{h}]$$

$$-\bar{\partial}(h\bar{h})\partial(h\bar{h}) + 2\frac{\bar{\partial}}{\partial_-}[h\partial_-\bar{h}]\partial(h\bar{h}) + 2\frac{\partial}{\partial_-}[\bar{h}\partial_-h]\bar{\partial}(h\bar{h})$$

$$+\frac{1}{\partial_-}[\partial_-h\bar{\partial}\bar{h} + \partial_-\bar{h}\bar{\partial}h]\partial(h\bar{h}) + \frac{1}{\partial_-}[\partial_-\bar{h}\partial h + \partial_-h\partial\bar{h}]\bar{\partial}(h\bar{h})$$

$$-2h\partial\bar{h}\frac{1}{\partial_-}[\partial_-h\bar{\partial}\bar{h}+\partial_-\bar{h}\bar{\partial}h] - 2\bar{h}\bar{\partial}h\frac{1}{\partial_-}[\partial_-h\partial\bar{h}+\partial_-\bar{h}\partial h] - 4h\partial\bar{h}\frac{\bar{\partial}}{\partial_-}[h\partial_-\bar{h}] - 4\bar{h}\bar{\partial}h\frac{\partial}{\partial_-}[\bar{h}\partial_-h]$$

$$+2h\partial\bar{h}\bar{\partial}(h\bar{h}) + 2\bar{h}\bar{\partial}h\partial(h\bar{h}) - \frac{2}{3}\partial(\bar{h}\bar{h}h)\bar{\partial}h - \frac{2}{3}\bar{\partial}(hh\bar{h})\partial\bar{h} - 2h\bar{h}\bar{\partial}h\partial\bar{h}$$